

Ayman Saber AI/ML Engineer

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github.com/AymanAyro • Military Status: Exempted

EDUCATION

Faculty of Engineering, Suez Canal University (SCU), Bachelor in Computer Engineering | 09/2019 – 07/2024
Cumulative Grade: 3.24/4 (Very Good) | Graduation Project Grade: Excellent Ismailia, Egypt

PROFESSIONAL EXPERIENCE

AI Engineer, Strategic Business Unit (SBU) 12/2024 – 02/2025
Alexandria, Egypt

- Designed and deployed end-to-end AI pipelines for enterprise clients using AWS.
- Developed workflows using LangChain, LLMs, & RAG dealing with database MongoDB.
- Applied LLMops, implementing features and monitoring chatbots using Node.Js & Restful APIs.

AI & ML Instructor (Part-Time)

AMIT Learning 10/2024 – 11/2024
Cairo, Egypt

- Trained 20+ students in Data Preprocessing, Cleaning, Feature Engineering, and ML Algorithms.

Engineering Camp (E. Camp) 01/2024 – present
Ismailia, Egypt

- Designed Curricula (Beginner: Intro to Python & Data Manipulation Frameworks, Intermediate: Data Preprocessing, Feature Engineering & ML Algorithms, Advanced: Deep Learning)
- Mentored 100+ students dealing ML & DL models; 95% achieved certification distinctively.

PROJECTS

Digital-Ḍād الرقمية (Graduation Project)

- Achieved **97.04% accuracy** (2.96% CER) via hybrid **ResNet152/BiLSTM (CNN/RNN) OCR** model with CTC Loss for text sequences trained on 108K+ Arabic samples (**KHATT + Arabic Alphabet datasets**)
- Engineered preprocessing pipeline (**DocTr/CRAFT**) with **98% segmentation accuracy**, preserving diacritics via Gaussian/median filters
- Deployed Flask API** backend + React.js frontend (exam grading/doc scanning features via Gemini API for **post-processing**)
- Published research** in Suez Canal University AI Conference and *International Journal of Telecommunications (IJT)*

Technologies: TensorFlow, OpenCV, React.js, Flask, Gemini API, Data Augmentation, LLMs, NLP

PCB Defects Detection

An object detection model able to identify the errors and defects made during the manufacturing process of PCBs, such as missing holes, shorts, missing bites, and other errors.

- YOLO Implementation:** Designed a PCB fault detection model using YOLO for accurate detection of defects like missing holes, mouse bites, and open circuits.
- Advanced Object Detection Techniques:** Leveraged cutting-edge object detection methods for precise identification of manufacturing defects.
- Integration-Ready Solution:** Developed the model for seamless integration with factory cameras, enabling automated real-time quality control.

Technologies: YOLO, OpenCV, Python, TensorFlow, Object Detection Techniques, Real-Time Video Processing

Article Processor

- Automated text-to-Markdown/JSON conversion using **Gemini API**, training models via **few-shot learning** for **context**.
- Enforced strict **metadata schemas** (Pydantic) to filter invalid/empty fields and ensure data integrity
- Integrated secure **API** workflows with error handling for **Gemini** interactions and **JSON** serialization
- Built **automated pipelines** (Pathlib) for **structured outputs** and file management

Technologies: Python, Gemini API, LangChain, Pydantic, JSON Schema, RESTful API Integration, Pathlib

ASL (Arabic Sign Language) Recognition

- Developed a **YOLOv8-based model** for real-time Arabic sign language detection from hand gestures, achieving **98.2% mAP50** - **95% F1-Score** for overall classes.
- Optimized training with a **cosine scheduler** and custom optimizer to enhance detection accuracy.
- Implemented **live inference** using **OpenCV** for real-time webcam processing with high-confidence predictions.

Technologies: YOLO, OpenCV, Python, Deep Learning, Real-Time Inference

Mental Health Chatbot - Pandora

- Developed AI chatbot using **BERT** and PyTorch for intent classification/sentiment analysis (Azure NLP), fine-tuned via Hugging Face Transformers
- Built Streamlit UI for real-time interactions + MLflow integration for tracking conversation metrics/model performance
- Engineered JSON intent workflows with tokenization/lemmatization pipelines and data visualizations (word clouds/analytics)

Technologies: Python, PyTorch, Transformers, Hugging Face, Streamlit, Azure Text Analytics, MLflow, NLTK, spaCy, and Matplotlib.

Car Price Prediction

- Developed** a predictive pipeline achieving a **94% R² score** for car price estimation, with **LightGBM** as the top performer.
- Cleaned and preprocessed** a dataset of **7,253 entries** by addressing missing values (Mileage, Engine, Power), removing outliers, and engineering features (e.g., **Extracting Features, Encoding, Dealing with Outliers**).

- **Designed and evaluated** five regression models—leveraging **MLflow** for experiment tracking and hyperparameter tuning.
 - **Delivered** a scalable solution for car price estimation, enabling informed decisions within a **2-day project timeframe**.
- Technologies:** Python, Pandas, Matplotlib, scikit-learn, XGBoost, LightGBM, MLflow

Iris Tracking Detection

- Achieved **0.019 MAE** via hybrid EfficientNetB3 architecture with custom spatial attention/CoordConv layers
 - Engineered adaptive augmentation (Albumentations) reducing training data needs by **83%** vs. baseline models
 - Implemented mixed-precision training: cyclical LR (1e-4→3e-3) + hybrid Huber/Log-Cosh loss with scheduling
 - Designed dual annotation system (LabelMe) for 145 high-res eye images: segmentation masks + sub-pixel keypoint vectors
- Technologies:** TensorFlow/Keras, OpenCV, Real-Time Inference, JSON, Tensorboard, Albumentations

VOLUNTEERING ACTIVITIES

IEEE Suez Canal University Student Branch (NGO), AI Technician, Leader, and Speaker (Volunteer)	08/2023 – 08/2024
• Led AI team of technicians to work on end-to-end project (Clustering Images Using CNNs) • Presented AI concepts (Machine Learning & Data Preprocessing) at workshops/seminars by hosting 2 bootcamps . • Provided technical expertise (AI models/tools) on 10+ projects (Regression, Classification, Recommendation Systems).	Ismailia, Egypt

PUBLICATIONS

- Ayman Saber, Ahmed Taha, & Khalid Abd-El Salam. **"A Comprehensive Approach to Arabic Handwriting Recognition: Deep Convolutional Networks and Bidirectional Recurrent Models for Arabic Scripts"**, *IJT*, 4(2), 1-11, 2024

SKILLS

Technical Expertise: Python • Scikit-Learn (Sklearn) • MLOPs • TensorFlow/Keras • PyTorch • YOLO • LangChain • Hugging Face • LLMs • LLMOps • Prompt Engineering • RAG • Computer Vision • Image Processing • NLP • NLTK • Generative AI (GenAI) • Exploratory Data Analysis (EDA) • Feature Engineering • Data Preprocessing • C++ • FastAPI • Docker • Streamlit • Algorithm Design • AWS • Azure • SQL/NoSQL • MLflow • APIs • Cloud Services • Model Deployment | **Soft Skills:** Researching • Documentation • Communication • Team Leadership • Agile Project Collaboration • Technical Mentorship

CERTIFICATES

Microsoft Machine Learning Engineer — Digital Egypt Pioneers Initiative
TOT (Training Of Trainers) Workshop — Creativa Ismailia, Aspire
Huawei HCIA-AI — Huawei ICT Academy, SCU

LANGUAGES

Arabic: Native • **English:** Advanced (C1)